



POLITECNICO
MILANO 1863

DEPARTMENT OF
MECHANICAL ENGINEERING

HILDDS PRESENTATION SUZUKA OVERTAKE

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AGENDA

- Introduction
- Objectives
- Driving scenario setup
- Offline and driving dynamic simulation
- Results
- Conclusions



INTRODUCTION

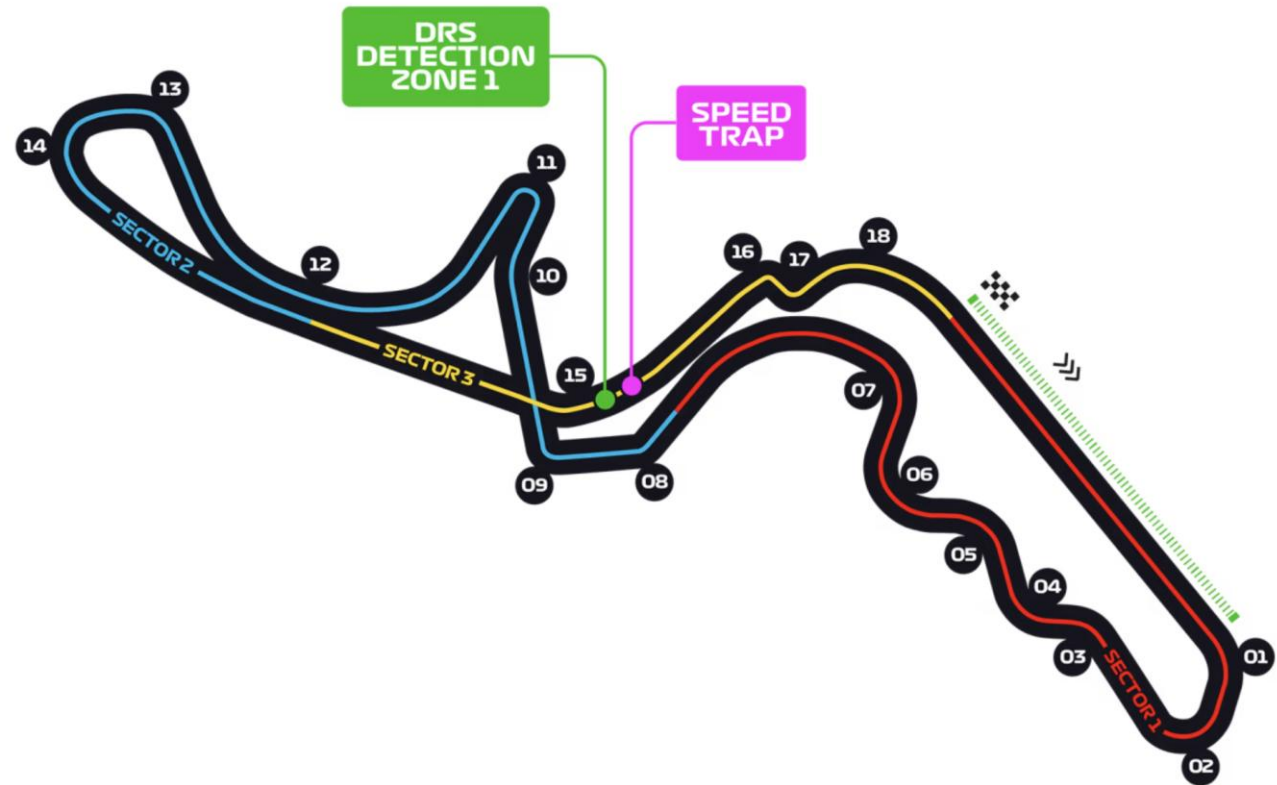
The aim of the project work was the design of a scenario to **simulate a driving maneuver** and analyze the **output data** comparing them to **drivers' feelings**.

STEPS:

- Define the **maneuver** and a vehicle
- Run **offline simulations**
- Design the **3D environment**
- Select **cueing** parameters to correctly run the maneuver at the simulator
- **Drive** the simulator
- Analyze the **data**

INTRODUCTION

- Which sector?
- Why Suzuka circuit?



OBJECTIVES

The main objective was to understand how different drivers would react to **unexpected slow cars in trajectory** while on a flying lap.

To analyze, during the **evasive maneuver**:

- Different **strategies**
- Followed **trajectories**
- Driver **feelings**
- Driving **styles**
- Induced **car behavior**

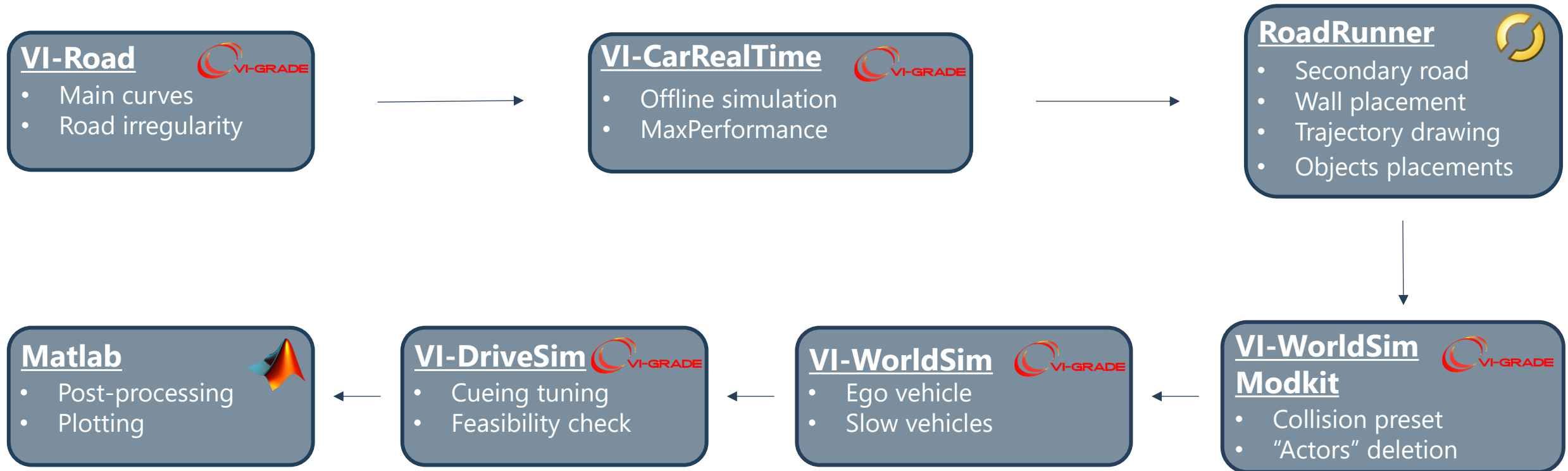
OBJECTIVES



Isack Hadjar and Ritomo Miyata incident in the Monaco tunnel in 2024 F2 qualifying

DRIVING SCENARIO SETUP

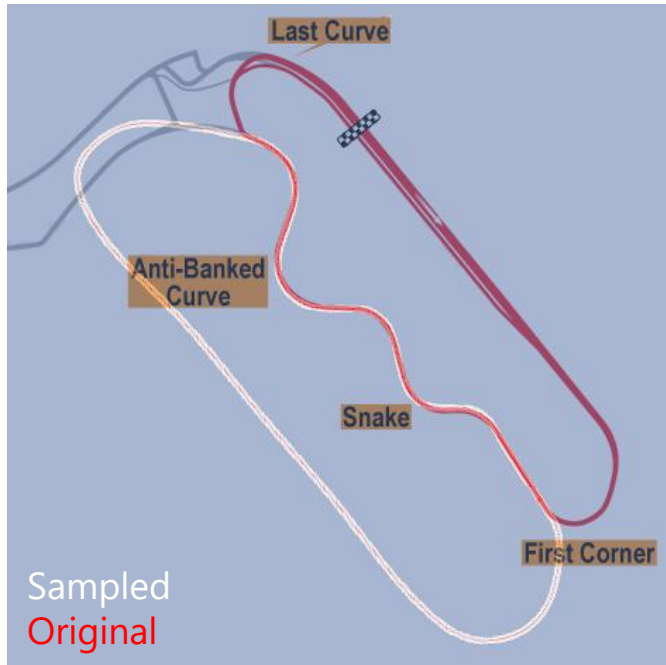
To create the driving scenario different **software** were used:



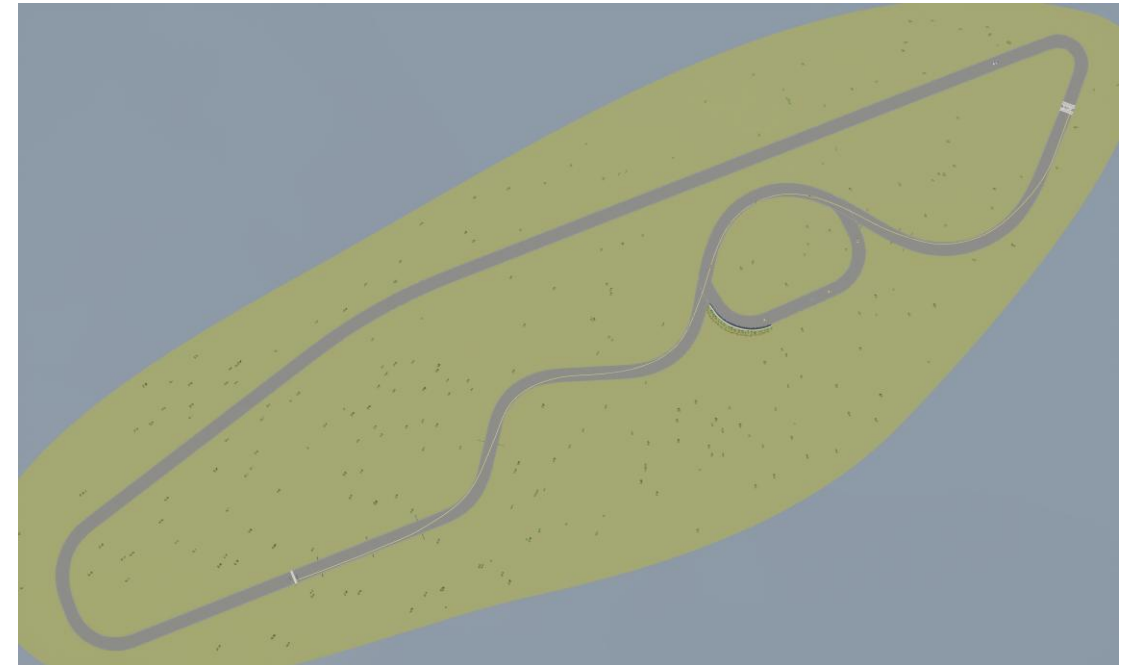
DRIVING SCENARIO SETUP

TRACK CREATION

The track was sampled using **Google Maps** satellite images, computing the **curvature radii** for each turn



Final track configuration, used in the simulator, with back road and the track for the slower cars, smoothed for the irregularity

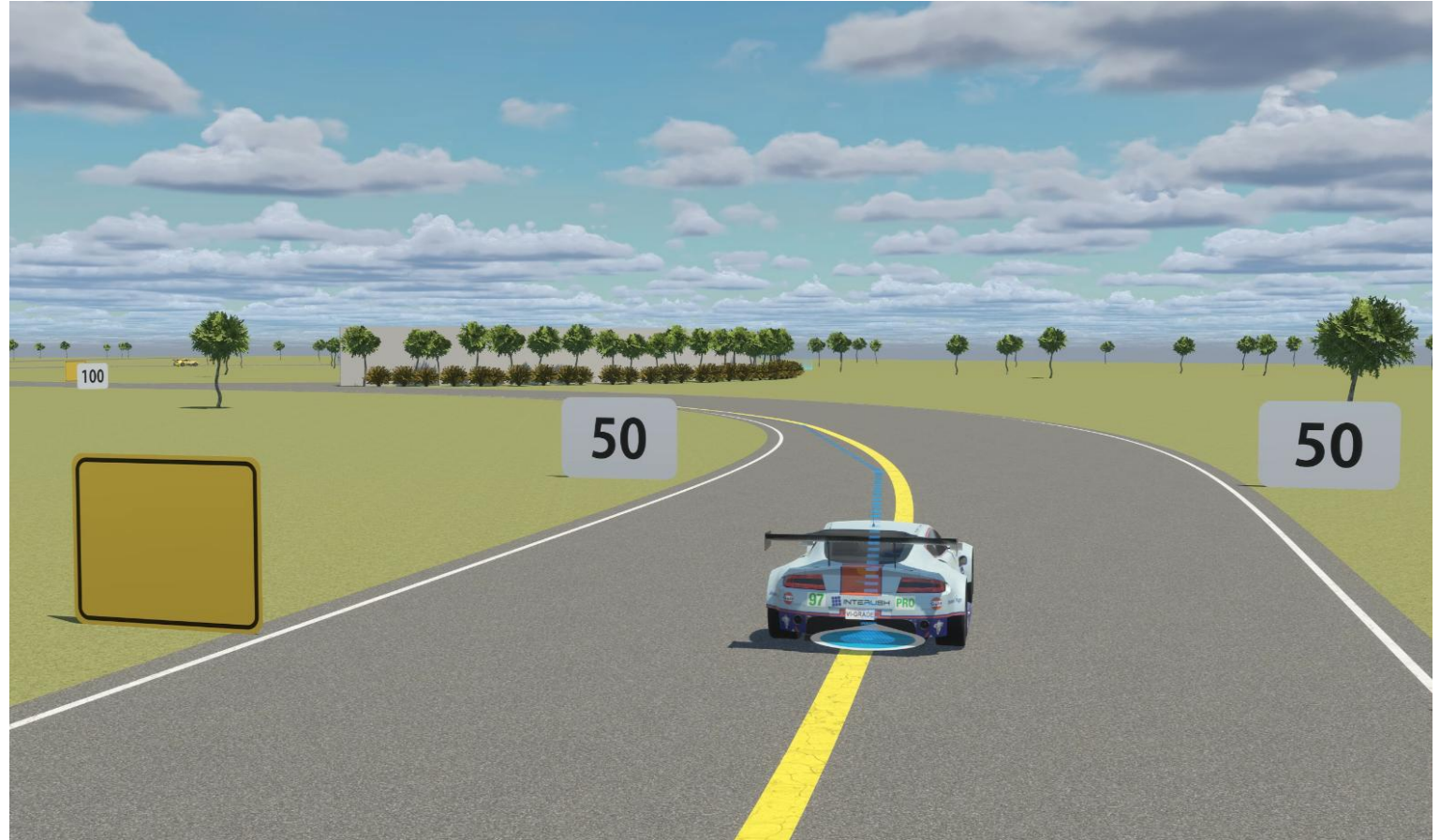


DRIVING SCENARIO SETUP

OBJECTS PLACEMENT

Several objects were placed to improve the scenario:

- **Ideal trajectory**
- **Braking points**
- **Turn distances**
- **Wall** and **vegetation**
- **Start** and **stop** signs



DRIVING SCENARIO SETUP

TRAFFIC IMPLEMENTATION

- **Slow vehicles** appeared **unexpectedly** in trajectory, disturbing the driver
- Each lap started at **different times**, to make the vehicles more **unpredictable**



OFFLINE AND DRIVING DYNAMIC SIMULATION

OFFLINE SIMULATION

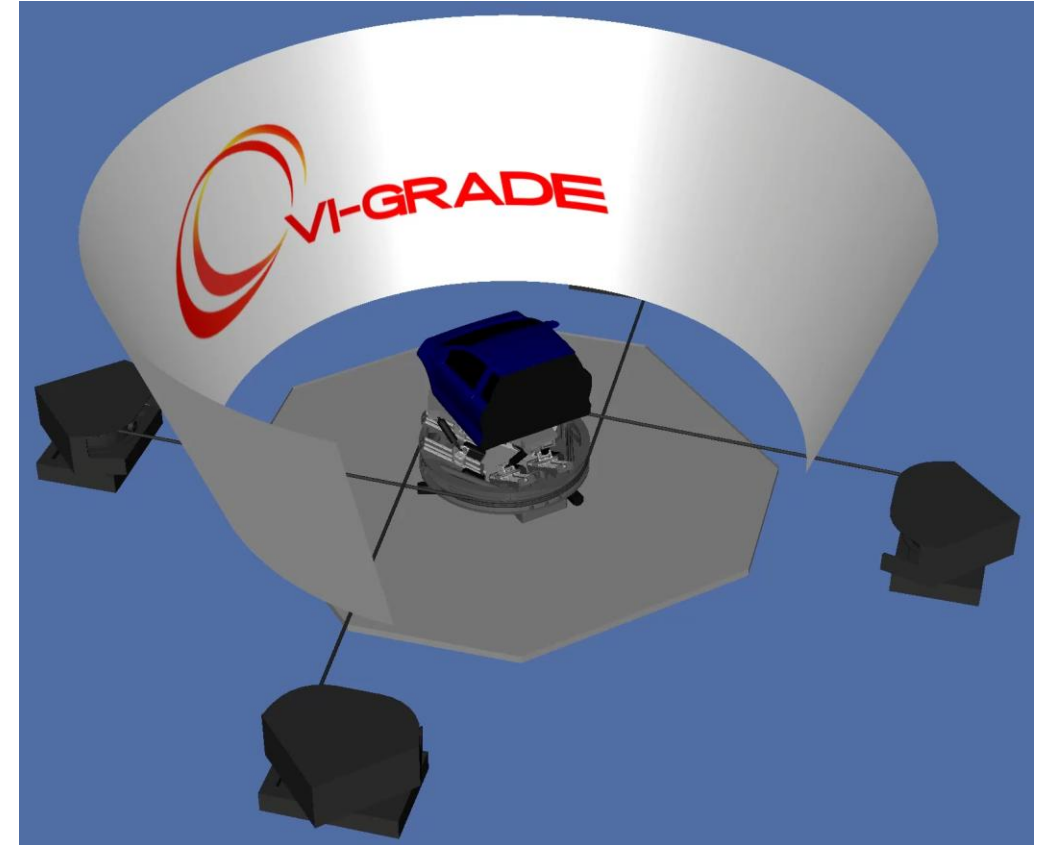
- Offline simulations allowed us to extract the **ideal trajectory** and **braking points** for drivers to follow
- Output data of the simulation were used as **reference**



OFFLINE AND DRIVING DYNAMIC SIMULATION

OFFLINE SIMULATION

Tuning the cueing algorithm resulted in the **reduction** of the **gains** concerning the **lateral accelerations**



OFFLINE AND DRIVING DYNAMIC SIMULATION

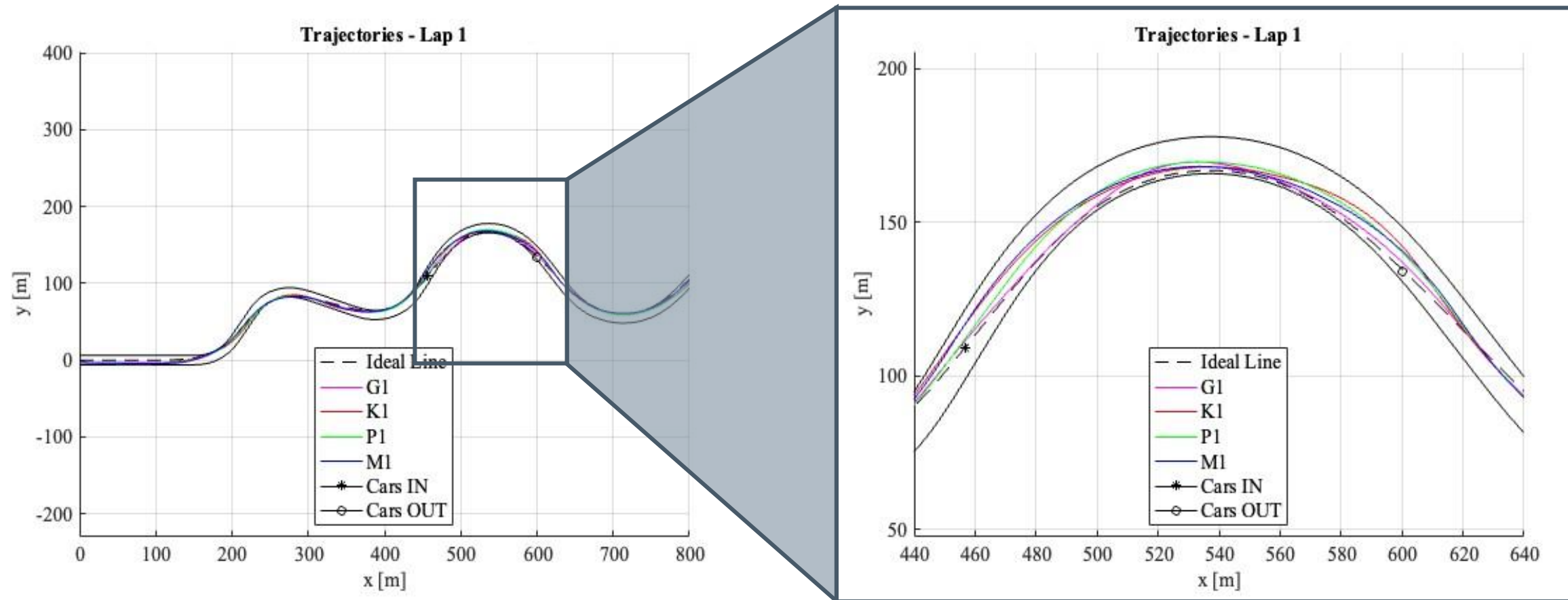
ONLINE SIMULATION

Actual **driving session** at the simulator



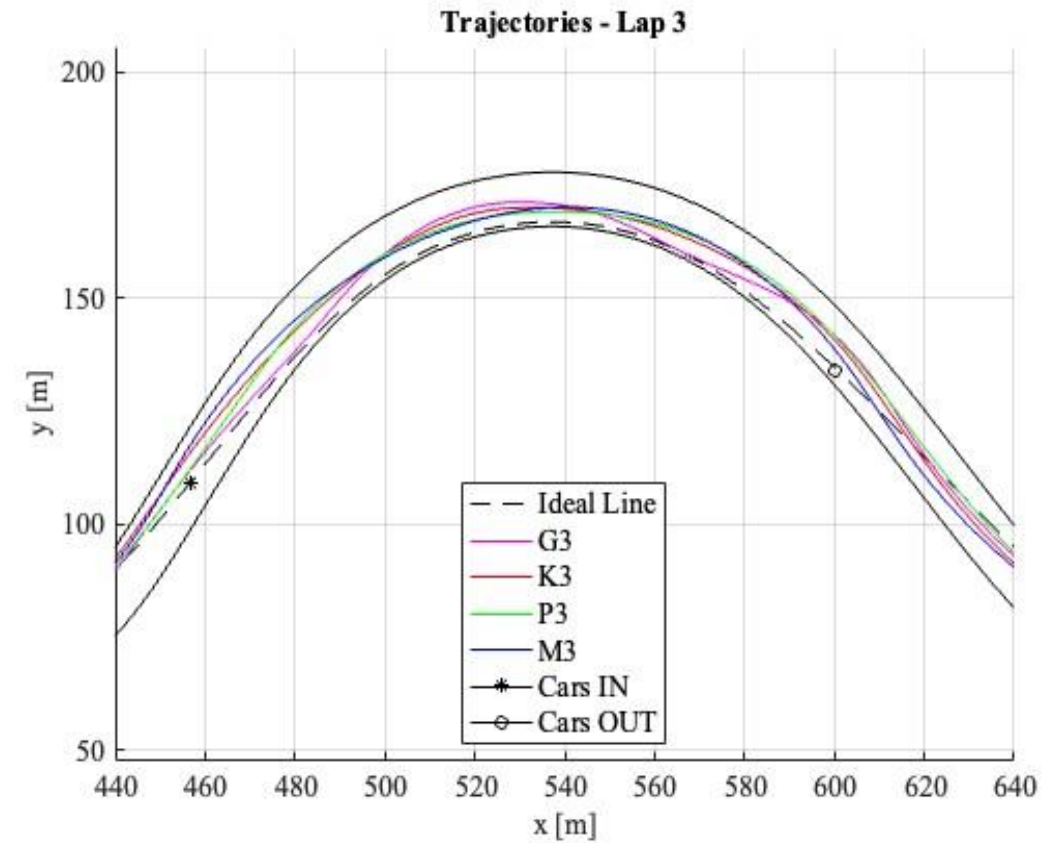
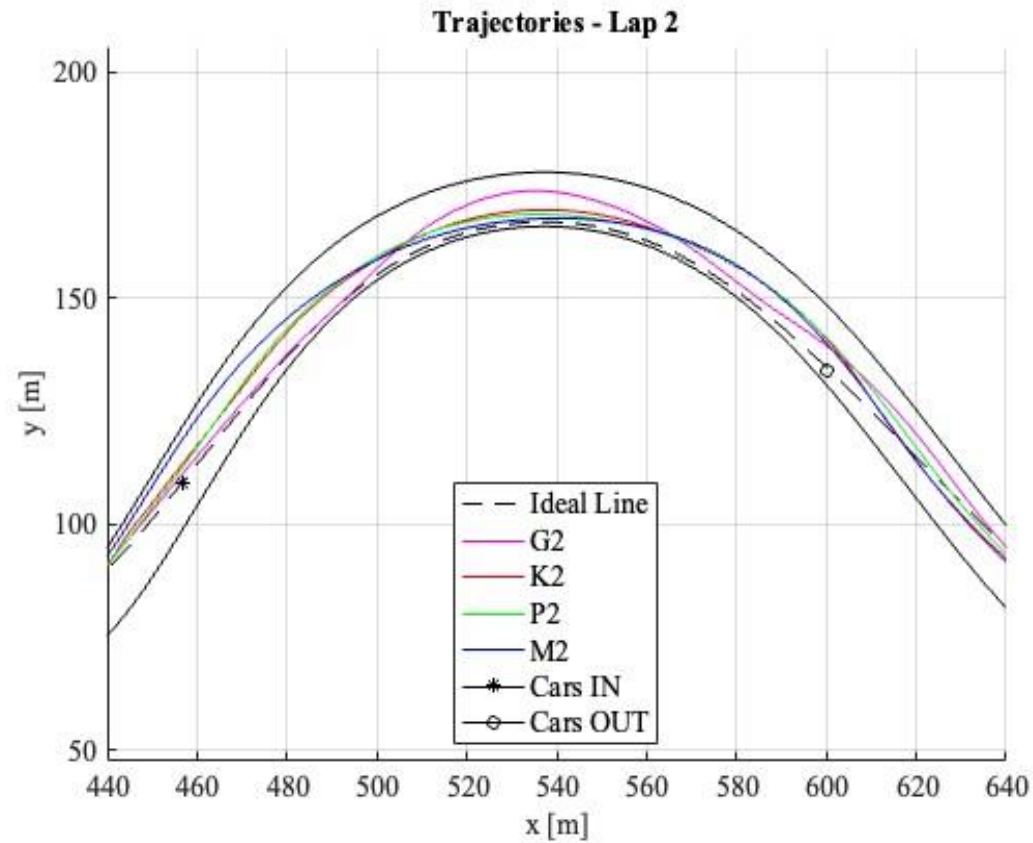
RESULTS

TRAJECTORIES



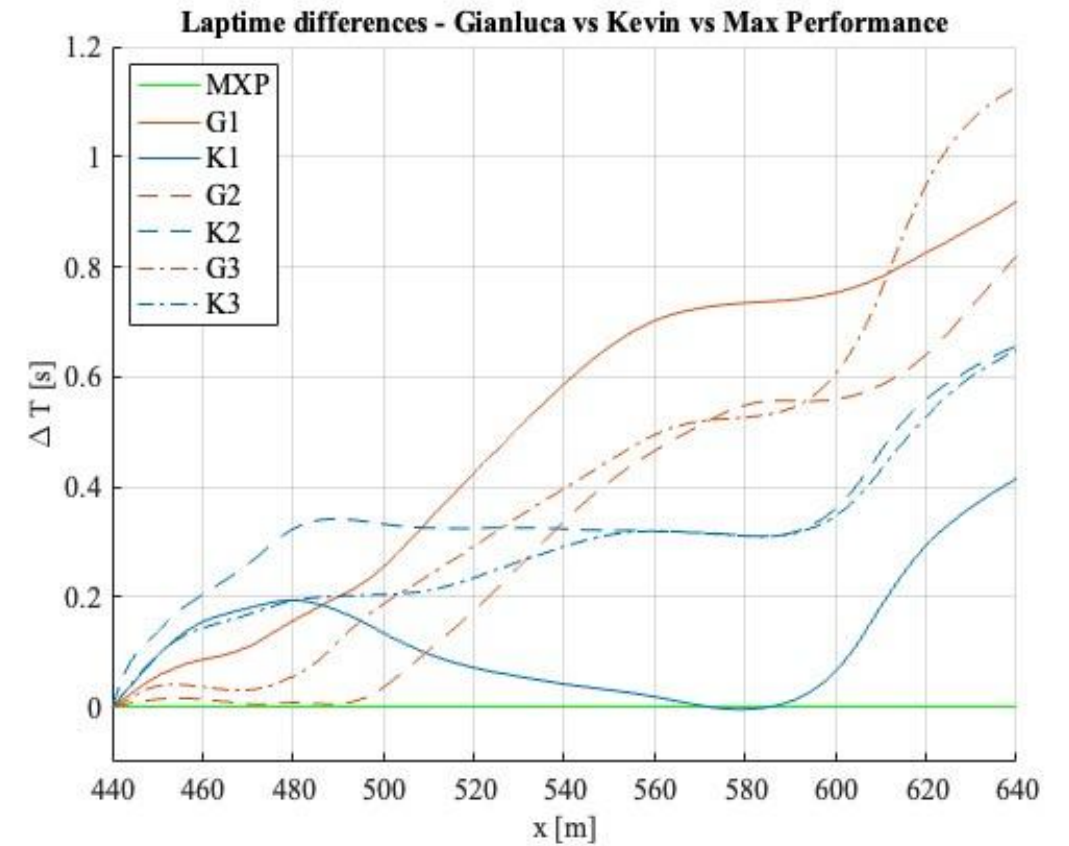
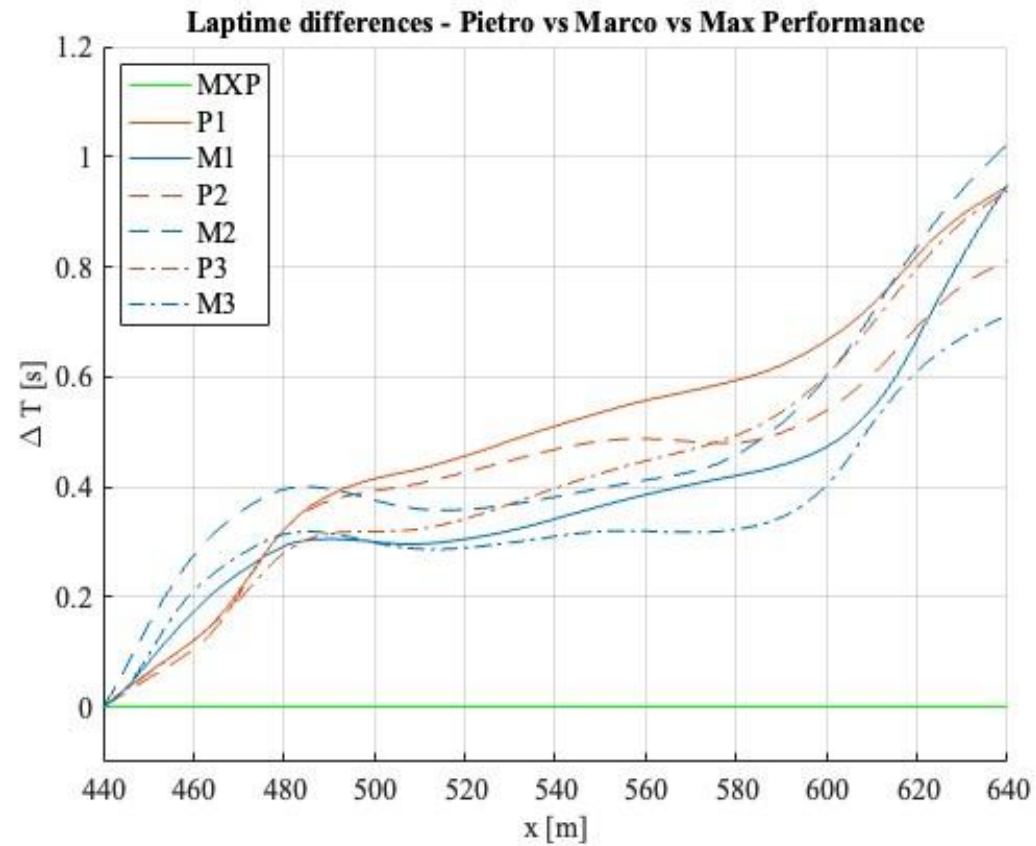
RESULTS

TRAJECTORIES



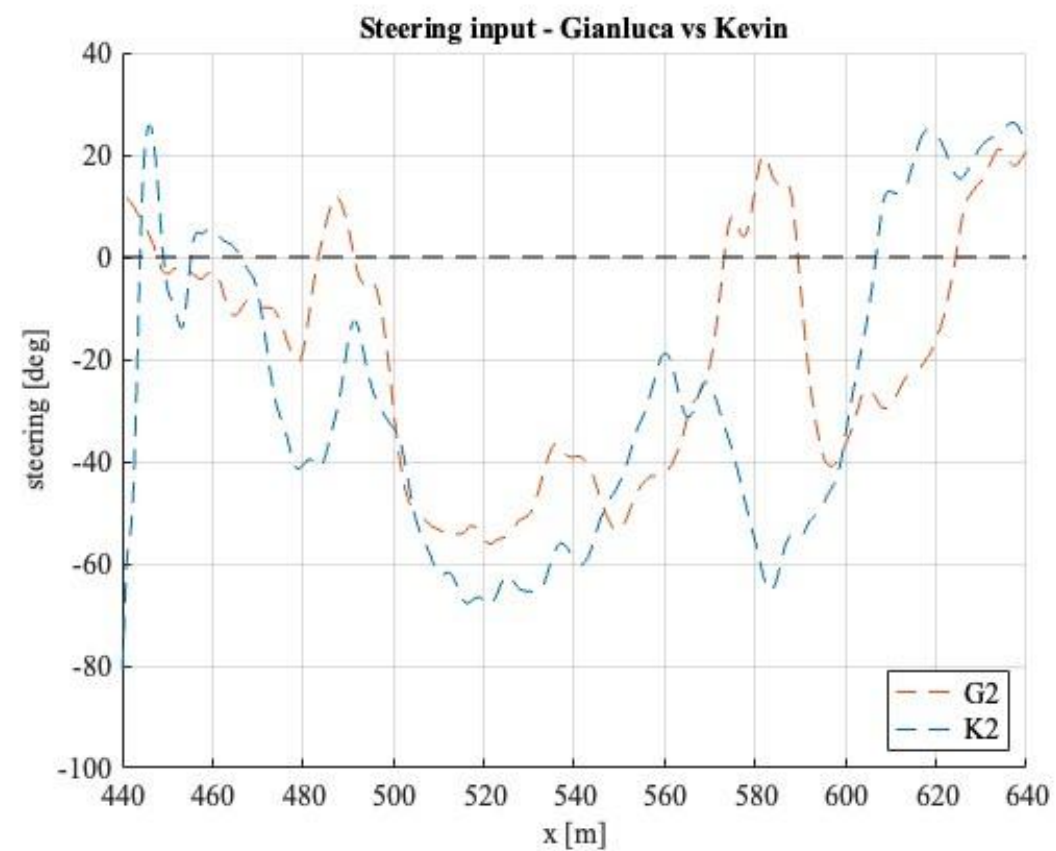
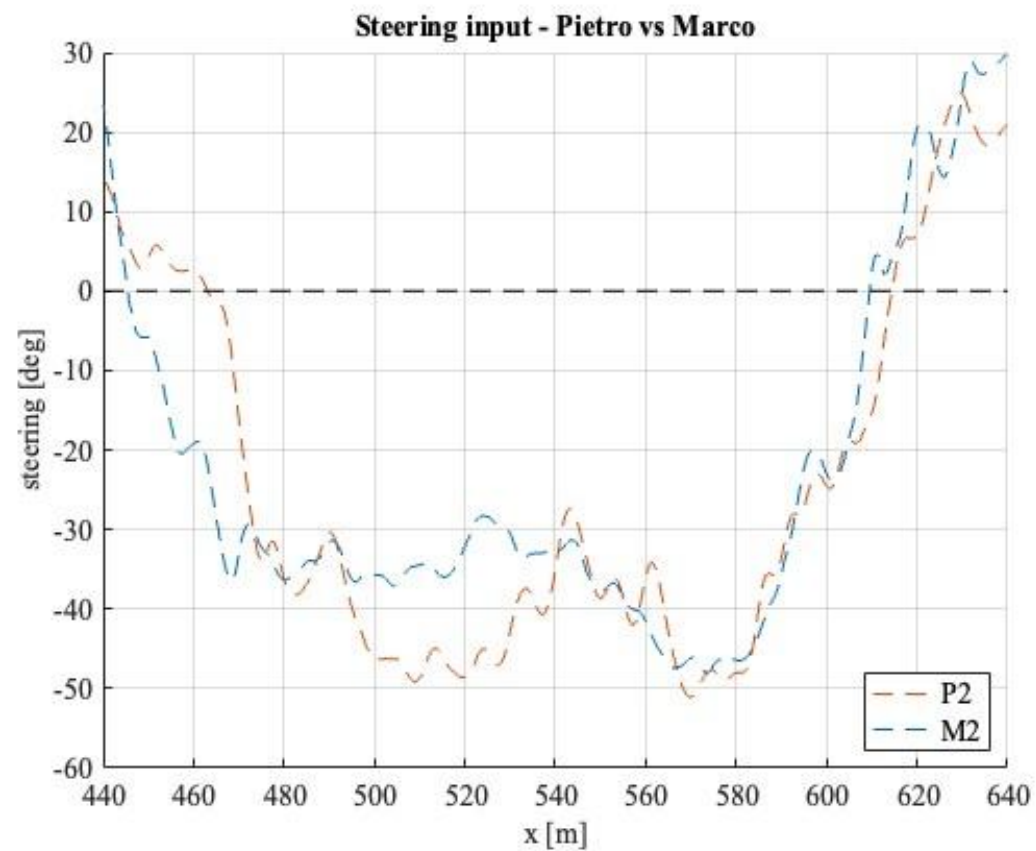
RESULTS

LAPTIME COMPARISON



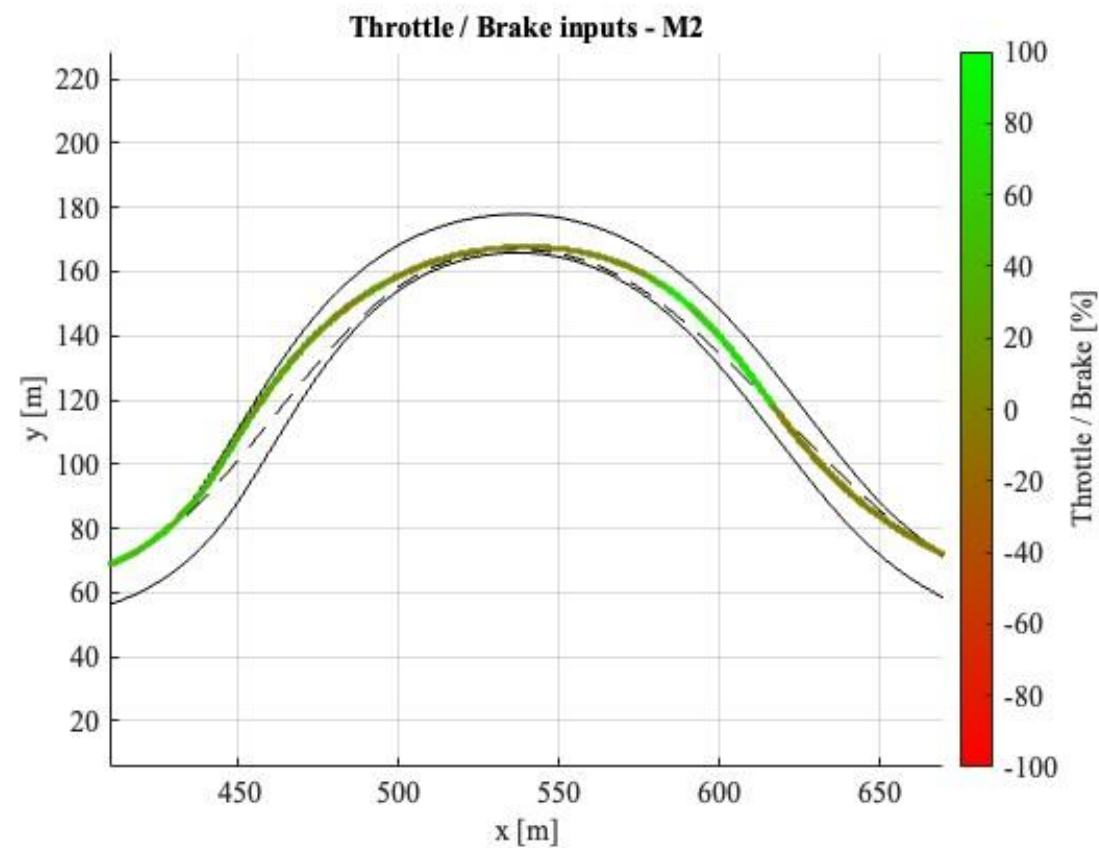
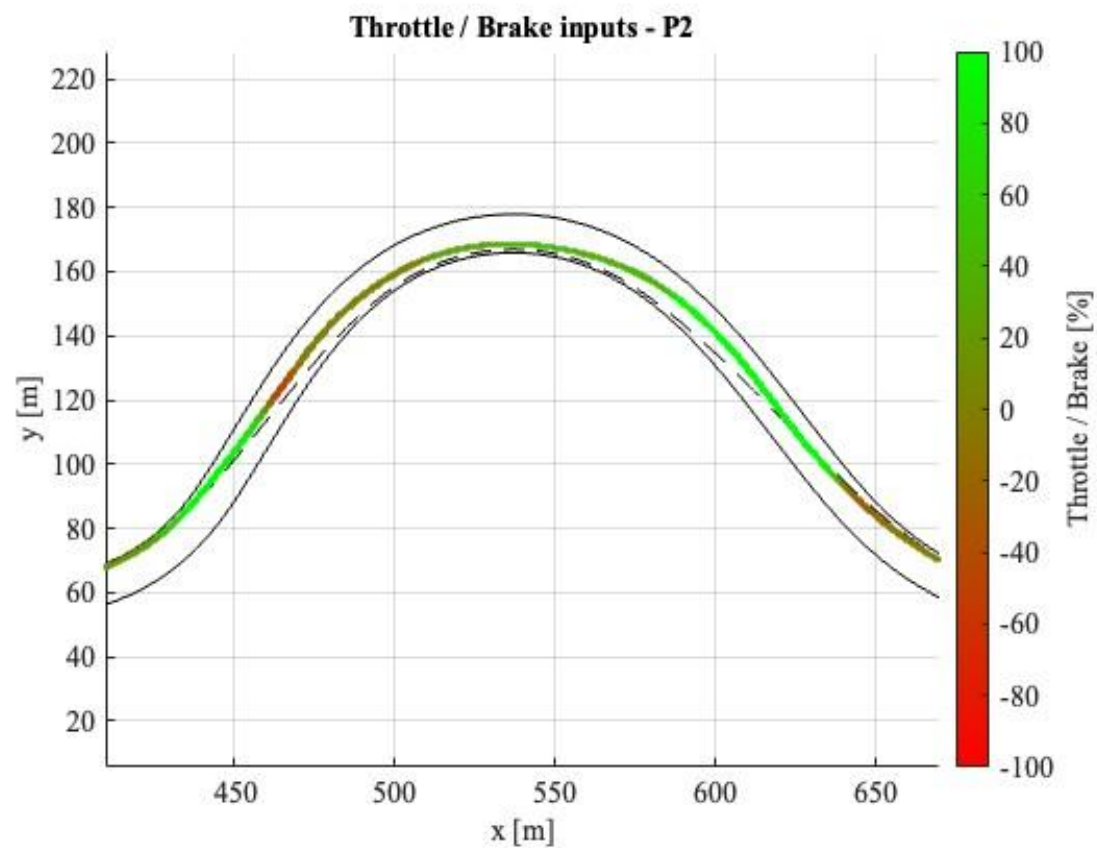
RESULTS

DRIVER INPUTS - STEERING



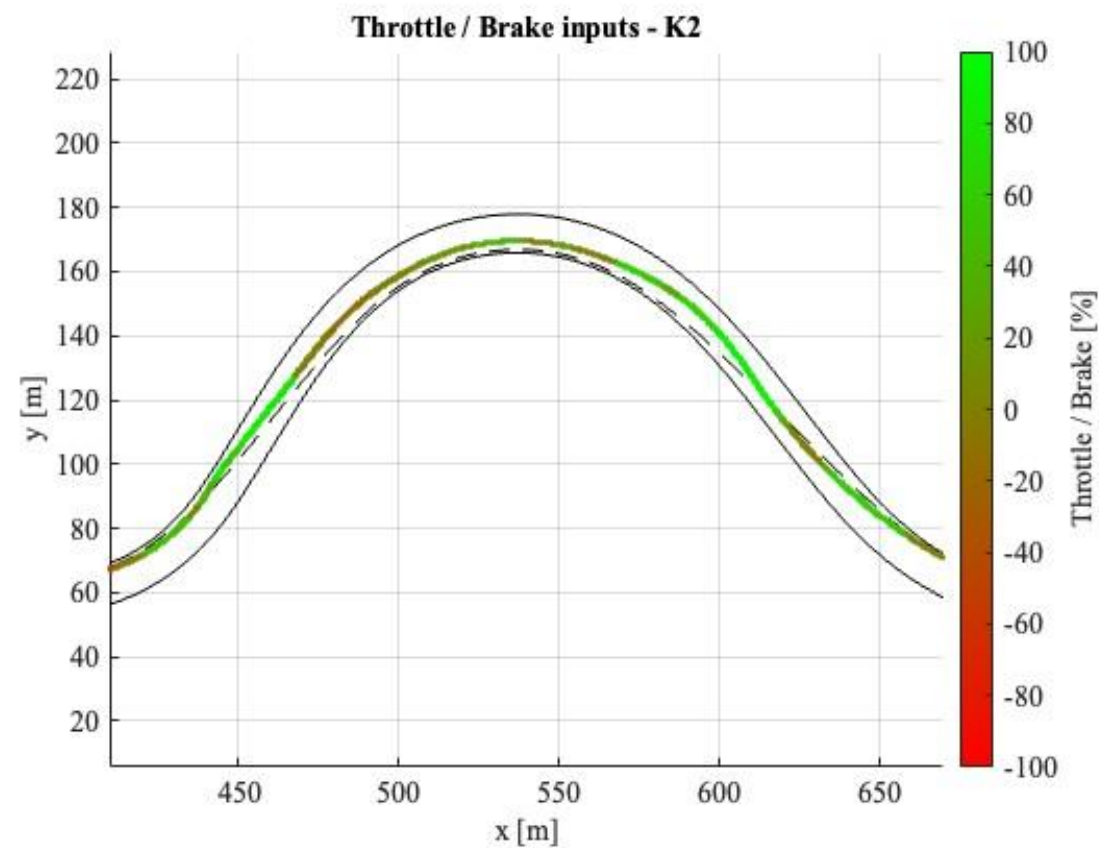
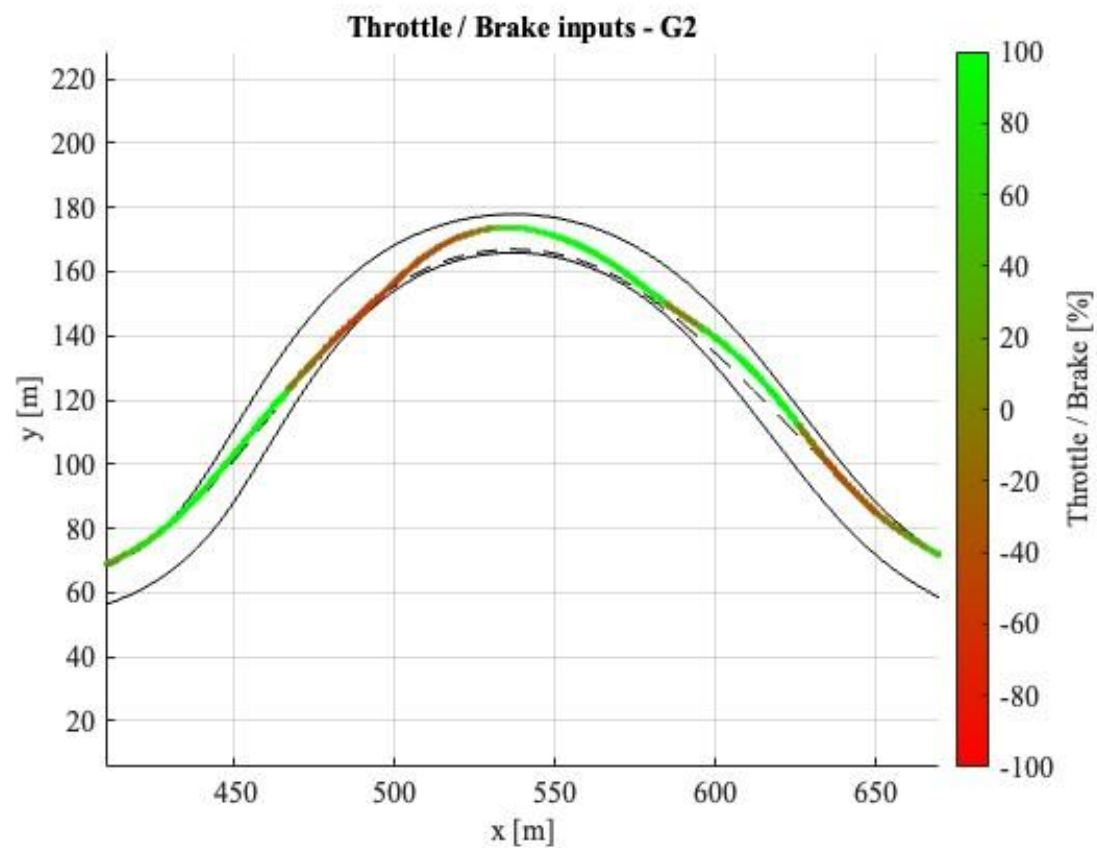
RESULTS

DRIVER INPUTS – THROTTLE AND BRAKE



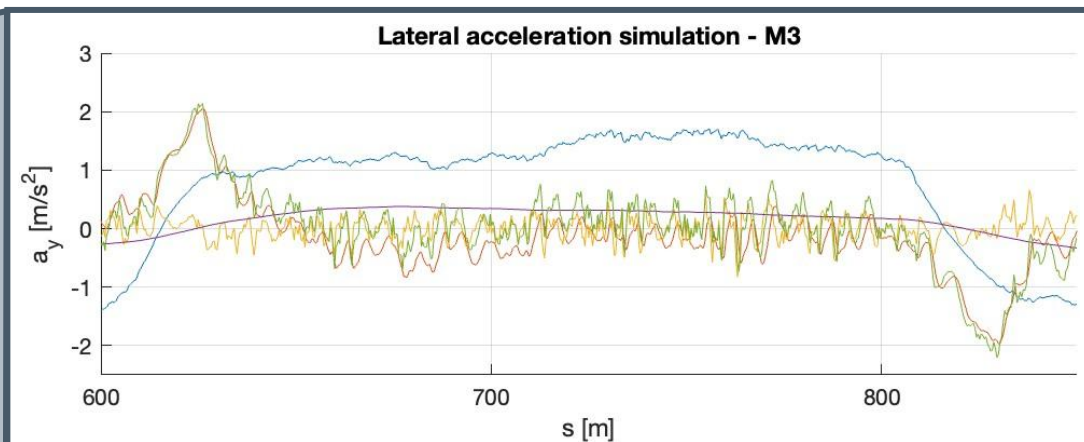
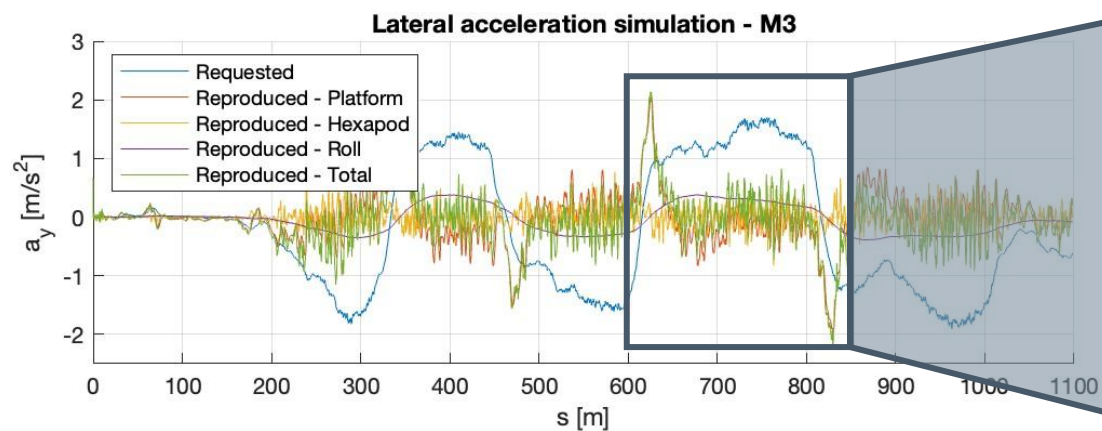
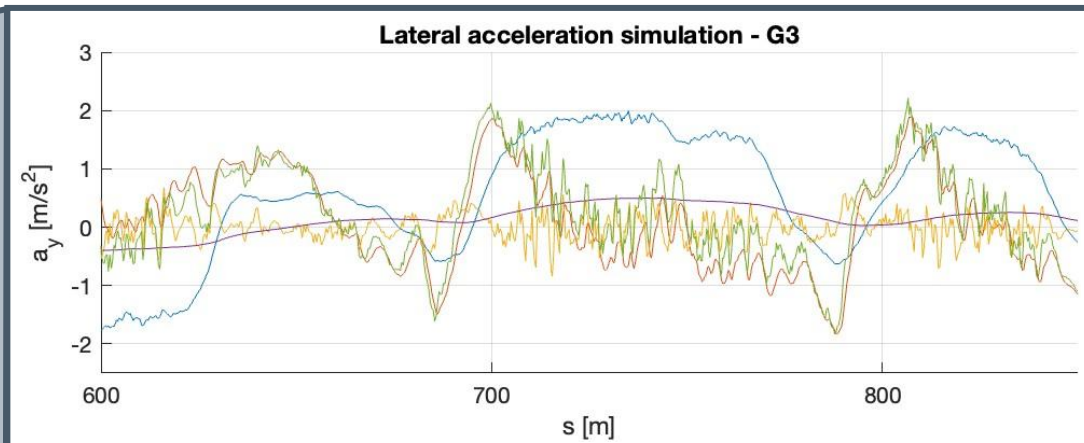
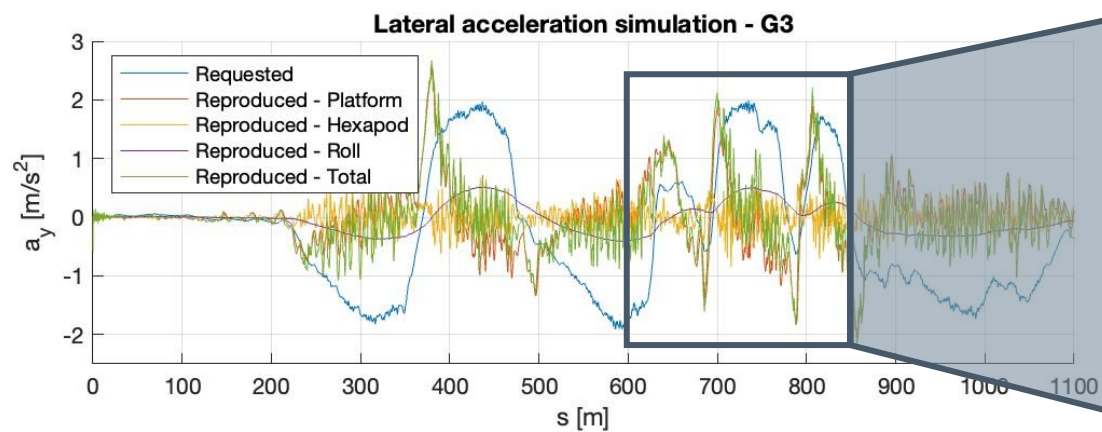
RESULTS

DRIVER INPUTS – THROTTLE AND BRAKE



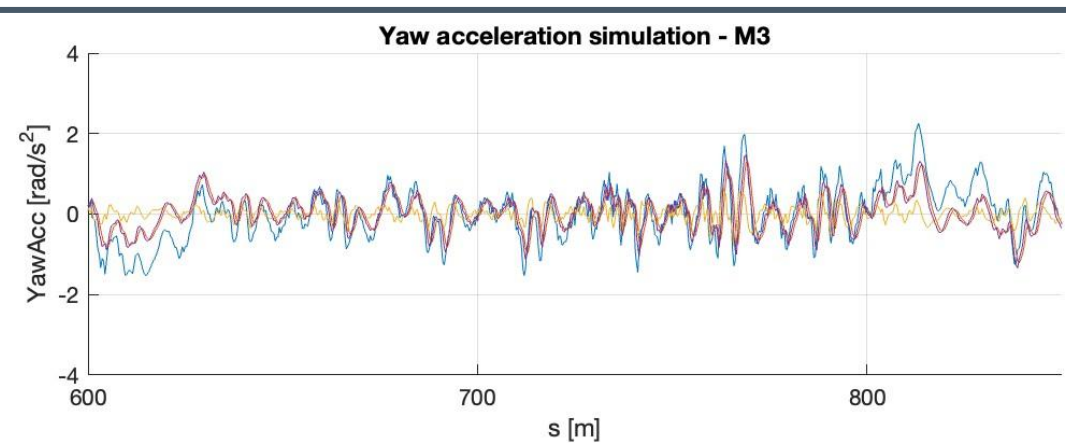
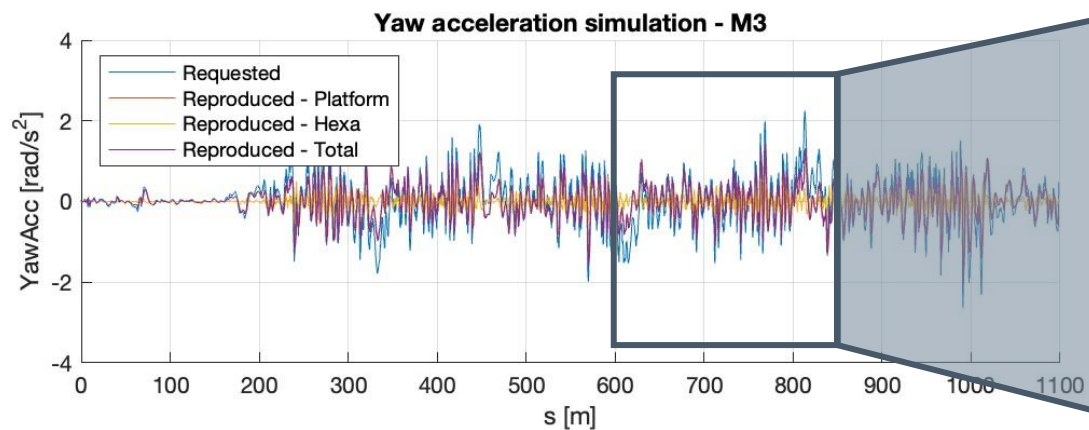
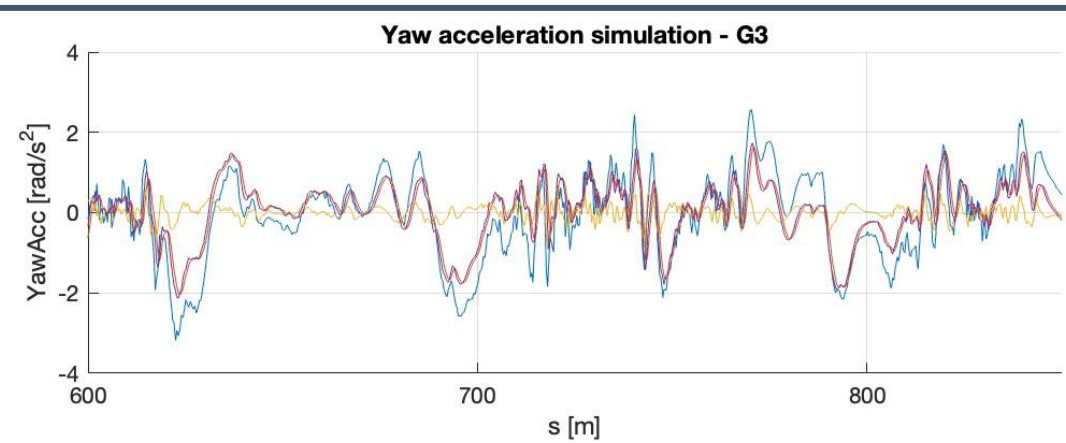
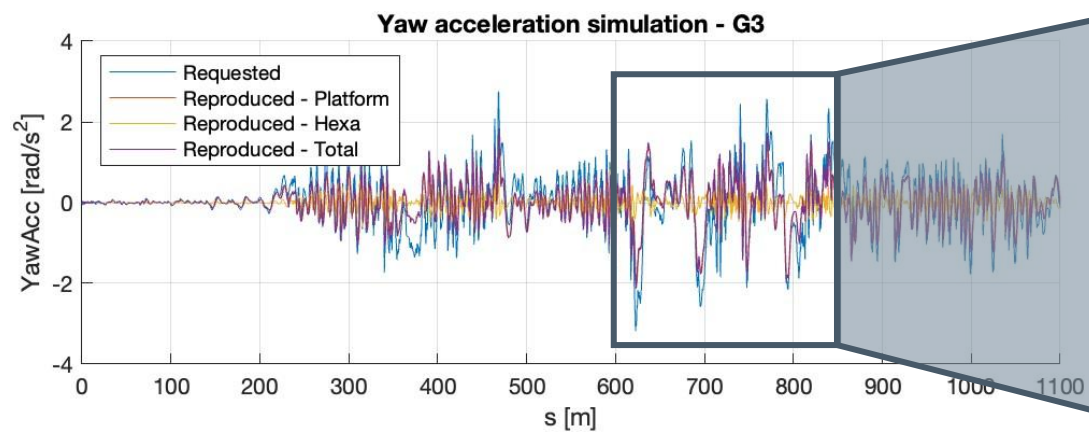
RESULTS

DRIVER FEELINGS – LATERAL ACCELERATIONS



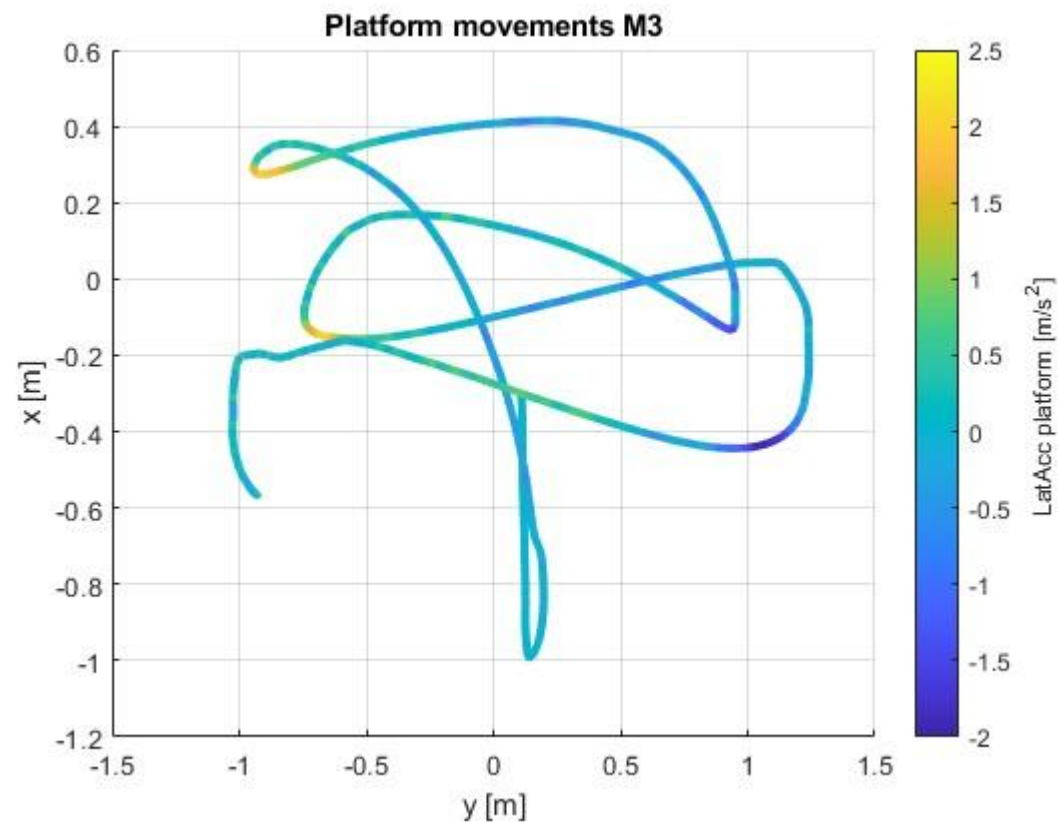
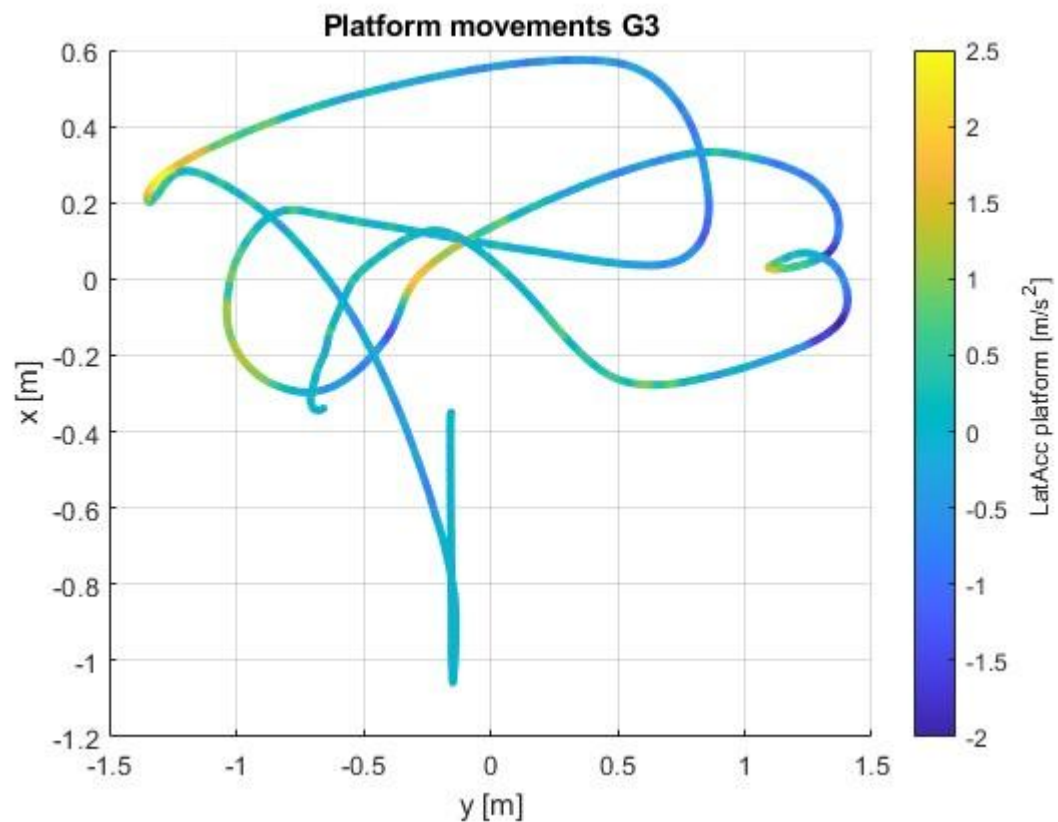
RESULTS

DRIVER FEELINGS – YAW ACCELERATIONS



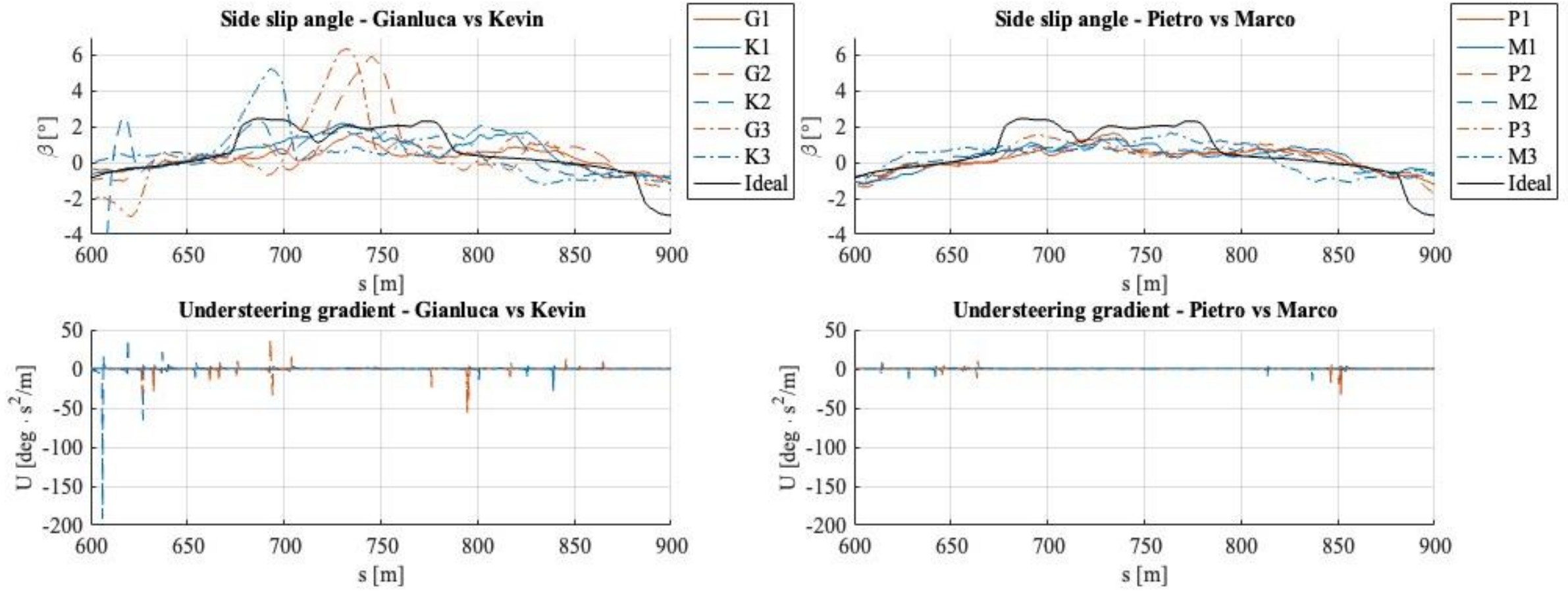
RESULTS

DRIVER FEELINGS – PLATFORM LIMITS



RESULTS

INDUCED CAR BEHAVIOR - STABILITY



GENERAL CONCLUSIONS

- **Effectiveness of the HIL Simulator:** Confirmed capability to faithfully replicate dynamic scenarios with unexpected traffic.
- **Robustness of the Cueing Model:** Lateral and yaw loads remained well within platform limits without saturation.
- **Validation of the Simulation Process:** Offline simulations provided a reliable reference for ideal trajectory and quantities.
- **Performance Differentiation:** Clear deviations in lap times and trajectories between runs with and without traffic.

CONCLUSIONS ON DRIVERS' FEELINGS

- **Dynamic Perception:** Individual variations in lateral and yaw accelerations influenced confidence and driving style.
- **Reaction Strategies:**
 - More aggressive driver (K): Maintained high throttle values and lost less time with respect to the ideal lap.
 - Intermediate approach (G): Followed ideal trajectory, heavier braking and higher distance due to the slow vehicle to overtake.
 - More conservative drivers (P,M): Applied smoother throttle and adopted wider trajectories.
- **Subjective Impact:** Qualitative feedback aligned with quantitative data, confirming the link between perceived comfort and performance.

CONCLUSIONS ON THE BEST APPROACH

- **Maintaining the Ideal Trajectory**
- **Brake Management**
- **Throttle Control**
- **Reacting to Unexpected Traffic**
- **Aggression–Safety Balance**
- **Top Performing Driver**



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